

An Entangled Affine Arboreal Image for $z \mapsto z^2$ at Basepoint 2

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Abstract

For the n th preimages of 2 under $z \mapsto z^2$, we determine the splitting field and Galois action uniformly for every $n \geq 3$. The pure radical and cyclotomic fields meet in exactly $\mathbb{Q}(\sqrt{2})$, and this shared quadratic field cuts the full affine group by the character equation $(-1)^b = (2/a)$. The resulting image has index two in $\text{AGL}_1(\mathbb{Z}/2^n\mathbb{Z})$, and adjacent-level restriction is surjective with kernel four. We further classify the complete fixed-root spectrum and prove that the odd primes for which $x^{2^n} = 2$ has a root have density $7/24 + 1/(3 \cdot 4^{n-1})$, tending to $7/24$; exactly one and four roots never occur. Finally, we identify the inverse-limit index-two 2-adic affine image, lock the all-iterate Frobenius-cycle dictionary, and independently exhaust 5,592,400 finite-image elements and 95,910 prime-level cells. The enumeration is a regression receipt, not a substitute for the proof. The fixed-root ledger does not supply an all-level primitive-cycle, orientation, phase, or monodromy atlas, so strict evaluator v0.2 records A1_WEAK and an exploratory overall status. The finite unitary structure supplies only A4_FORMAL_HINT.

Keywords: Kummer dynamics; arboreal Galois representations; 2-adic affine groups; Frobenius permutations; exact computation; finite Koopman operators; Route-A evaluation

中文摘要

本文考察平方映射在基点二之上的逐层原像域。第 n 层的根可用 2^n 次单位根编号；纯根式域与分圆域恰好共享 $\mathbb{Q}(\sqrt{2})$ ，这二个二次交叠迫使仿射参数满足模八的二次特征等式。由此得到指数二的精确Galois像，并证明相邻层限制满射且核的阶为四。进一步把Frobenius不动根化为一个二进线性同余。完整计数只允许零、二以及从八起的二次幂根数，特别排除一根与四根情形。结合Chebotarev定理，所得素数条件自然密度为 $7/24 + 1/(3 \cdot 4^{n-1})$ ，并单调趋于 $7/24$ 。逆极限给出一个闭的指数二二进仿射群；有限层群枚举和素数网格仅作实现回归。邻近基点三、未切口的全仿射群及合数标签分解构成三类A0对照；其中素数幂保留为Frobenius重复，只有混合合数被排除为单素数标签。现有资料没有完备的本原轨道、重复、定向、相位或稳定性图册。故评估记为A1_WEAK与探索状态，有限么正结构只记为A4_FORMAL_HINT，且不推出目标Euler乘积、零点对应或Hilbert-Pólya算子。
关键词：库默尔动力学；二进仿射群；Frobenius置换；精确枚举；有限Koopman算子；路线A评估；本原轨道障碍

1 Preimage fields and the shared quadratic field

Let $m = 2^n$, choose compatible elements

$$\alpha_n = 2^{1/m}, \quad \zeta_m = e^{2\pi i/m},$$

and put

$$R_n = \{\zeta_m^j \alpha_n : j \in \mathbb{Z}/m\mathbb{Z}\}, \quad K_n = \mathbb{Q}(R_n) = \mathbb{Q}(\alpha_n, \zeta_m).$$

These are the level- n vertices and splitting fields of the preimage tree of 2 for $z \mapsto z^2$; see Jones [1] for the general arboreal framework. The present monomial tower is solvable, but it is not the generic full-affine Kummer tower: $\alpha_n^{m/2} = \sqrt{2}$ already lies in the cyclotomic field.

Lemma 1 (Cyclotomic nonsquare). *For every $n \geq 3$, $\sqrt{2}$ lies in $C_n := \mathbb{Q}(\zeta_m)$ but is not a square in C_n .*

Proof. Using the standard cyclotomic description [3], because $\zeta_8 = \zeta_m^{2^{n-3}}$, $\sqrt{2} = \zeta_8 + \zeta_8^{-1} \in C_n$. If $y^2 = \sqrt{2}$ in C_n , then $y = \pm 2^{1/4}$, so $\mathbb{Q}(2^{1/4})$ would be an intermediate field of the abelian Galois extension $C_n/\mathbb{Q}$. Every such intermediate extension is Galois. This is impossible: $X^4 - 2$ is Eisenstein at 2, while the real field $\mathbb{Q}(2^{1/4})$ does not contain its conjugate $i2^{1/4}$. $\square$

Theorem 1 (Radical–cyclotomic intersection). *For all $n \geq 3$,*

$$\mathbb{Q}(\alpha_n) \cap \mathbb{Q}(\zeta_m) = \mathbb{Q}(\sqrt{2}), \quad [K_n : \mathbb{Q}] = 2^{2n-2}.$$

Proof. Over C_n , α_n satisfies $X^{m/2} - \sqrt{2}$. Capelli's binomial criterion [2] reduces irreducibility to excluding a square and, when $4 \mid m/2$, the exceptional shape $\sqrt{2} = -4c^4$. The first is Lemma 1. The second would also make $\sqrt{2}$ a square, namely $(2ic^2)^2$, because $i \in C_n$. Hence $[K_n : C_n] = m/2$, and

$$[K_n : \mathbb{Q}] = (m/2)\varphi(m) = 2^{2n-2}.$$

Eisenstein gives $[\mathbb{Q}(\alpha_n) : \mathbb{Q}] = m$. The compositum formula now gives intersection degree two; the contained field $\mathbb{Q}(\sqrt{2})$ is therefore the whole intersection. $\square$

2 Exact affine image and compatible restrictions

For odd a , let

$$\chi_2(a) = \left(\frac{2}{a}\right) = (-1)^{(a^2-1)/8},$$

viewed modulo 8. Define

$$H_n = \{(a, b) \in (\mathbb{Z}/m\mathbb{Z})^\times \ltimes \mathbb{Z}/m\mathbb{Z} : (-1)^b = \chi_2(a)\},$$

where (a, b) sends j to $aj + b$.

Theorem 2 (Exact image and restriction). *The action on R_n identifies $\text{Gal}(K_n/\mathbb{Q})$ with H_n . This group has order 2^{2n-2} and index two in $\text{AGL}_1(\mathbb{Z}/m\mathbb{Z})$. Under compatible choices $\alpha_n = \alpha_{n+1}^2$ and $\zeta_{2^n} = \zeta_{2^{n+1}}^2$, restriction is*

$$H_{n+1} \longrightarrow H_n, \quad (a, b) \longmapsto (a, b) \pmod{2^n}.$$

It is surjective and its kernel has order four.

Proof. Every automorphism has unique parameters $\sigma(\zeta_m) = \zeta_m^a$ and $\sigma(\alpha_n) = \zeta_m^b \alpha_n$, hence acts by $j \mapsto aj + b$. The two expressions for the shared square root give

$$\sigma(\sqrt{2}) = (-1)^b \sqrt{2} = (\zeta_8^a + \zeta_8^{-a}) = \chi_2(a) \sqrt{2}.$$

Thus the image lies in H_n . There are $\varphi(m)$ choices of a and $m/2$ choices of the forced parity of b , so $|H_n| = 2^{2n-2} = [K_n : \mathbb{Q}]$. The faithful inclusion is an equality.

Squaring the radical and root-of-unity formulas proves coordinate reduction. Each coordinate has two lifts. Reduction preserves a modulo 8 and the parity of b , so all four lifts satisfy the same character equation. This proves surjectivity and the kernel count. $\square$

Round 0 advance: field and compatible-image closure. The intersection, every finite image, and every adjacent restriction map are proved for all levels $n \geq 3$; no finite-level extrapolation occurs.

3 The complete fixed-root spectrum

For $(a, b) \in H_n$, fixed roots correspond to solutions of

$$(a-1)j \equiv -b \pmod{2^n}. \tag{1}$$

Theorem 3 (Fixed-root law). *The exact numbers of elements of H_n having a prescribed number of fixed level- n roots are*

<i>fixed roots</i>	<i>number of elements</i>
2	2^{2n-4}
$2^k, 3 \leq k < n$	$2^{2n-2k-1}$
2^n	1
0	$2^{2n-2} - 2^{2n-4} - \sum_{k=3}^{n-1} 2^{2n-2k-1} - 1$

No element has exactly one or four fixed roots.

Proof. Equation (1) has $g = \gcd(a-1, 2^n)$ solutions exactly when $g \mid b$. Since a is odd, $g \geq 2$, excluding one root. If $a \equiv 7 \pmod{8}$, then $\chi_2(a) = 1$, b is even, and $v_2(a-1) = 1$. The $2^{n-3} \cdot 2^{n-1} = 2^{2n-4}$ such pairs have two fixed roots.

For $3 \leq k < n$, there are 2^{n-k-1} residues with $v_2(a-1) = k$ and 2^{n-k} admissible b divisible by 2^k , giving the second row. When $a = 1$, only $b = 0$ fixes all roots. Finally, $v_2(a-1) = 2$ means $a \equiv 5 \pmod{8}$, so $\chi_2(a) = -1$ forces b odd; then $4 \nmid b$, excluding four fixed roots. The complement has none. $\square$

4 Chebotarev root densities

All odd primes are unramified in K_n , since the polynomial discriminant has no odd prime divisor. Its reduction is separable, and $x^{2^n} = 2$ has a root over $\mathbb{F}_p$ exactly when arithmetic Frobenius fixes a member of R_n .

Theorem 4 (Root-prime density). *For every $n \geq 3$, the odd primes at which $x^{2^n} = 2$ has a root have natural density*

$$\delta_n = \frac{7}{24} + \frac{1}{3 \cdot 4^{n-1}}. \quad (2)$$

The sequence decreases to $7/24$. Each nonzero root multiplicity in Theorem 3 separately has the density obtained by dividing its group count by 2^{2n-2} .

Proof. The fixed-point property and fixed-point count are conjugacy invariant. Chebotarev [4] therefore identifies the requested densities with the corresponding proportions in H_n . The positive fraction is

$$\frac{1}{4} + \sum_{k=3}^{n-1} 2^{1-2k} + 2^{-2n+2} = \frac{7}{24} + \frac{1}{3 \cdot 4^{n-1}}.$$

The limit and monotonicity are immediate. $\square$

Round 1 advance: fixed-root and density closure. The complete Frobenius fixed-root spectrum, the impossible multiplicities 1 and 4, every multiplicity density, and the limit $7/24$ are added as all-level theorems.

5 Inverse limit and all-iterate Frobenius cycles

The character condition is compatible because it depends only on $a \pmod{8}$ and $b \pmod{2}$.

Proposition 1 (Infinite arboreal image). *The inverse-limit Galois image is*

$$H_\infty = \{(a, b) \in \mathbb{Z}_2^\times \times \mathbb{Z}_2 : (-1)^{b \bmod 2} = \chi_2(a)\}.$$

It is the kernel of a continuous surjective character of the 2-adic affine group and hence is closed of index two.

Proof. Theorem 2 supplies a surjective inverse system. Under $(a, b)(c, d) = (ac, ad + b)$, oddness of a gives

$$\chi_2(ac)(-1)^{ad+b} = \chi_2(a)(-1)^b \chi_2(c)(-1)^d.$$

Thus the defining expression is a continuous character. It takes both signs and its finite quotients are exactly the H_n . $\square$

For an odd prime p , the degrees of the irreducible factors of $x^{2^n} - 2 \pmod{p}$ are the cycle lengths of arithmetic Frobenius. If $N_{p,n}(r)$ is the number of fixed roots of its r th power and $E_{p,n}(d)$ counts points of exact period d , then the inherited finite- permutation identities are

$$N_{p,n}(r) = \sum_{e \mid r} E_{p,n}(e), \quad E_{p,n}(d) = \sum_{e \mid d} \mu(e) N_{p,n}(d/e).$$

HCS-C12A already owns this universal mechanism. C374 owns its specific compatible Kummer image and the closed fixed-root/density atlas above.

6 Exact evidence and independent validation

The canonical ledger exhausts every element at levels $3 \leq n \leq 12$. It also checks every odd prime $p \leq 100000$ by the cyclic-group criterion: with $d = \gcd(2^n, p-1)$, the root count is d precisely when $2^{(p-1)/d} = 1 \pmod{p}$, and otherwise zero.

n	$ H_n $	exact δ_n	root primes $\leq 10^5$	odd primes tested
3	16	5/16	2,995	9,591
4	64	19/64	2,843	9,591
5	256	75/256	2,808	9,591
6	1,024	299/1024	2,803	9,591
8	16,384	4779/16384	2,799	9,591
10	262,144	76459/262144	2,798	9,591
12	4,194,304	1223339/4194304	2,797	9,591

Table 1: The exact density column is proved; the finite prime column is a regression receipt and is not used to infer it.

Across all ten levels, the exhaustive group lane checks 5,592,400 pairs; the prime lane checks 95,910 cells. A non-importing checker recomputes all counts, a SymPy lane verifies independent identities, two isolated builds must agree byte for byte, and repaired-hash mutations must be rejected.

7 Finite Koopman realization and Route-A boundary

Each H_n acts canonically on $\ell^2(R_n)$ by permuting the orthonormal point basis, hence by real permutation unitaries with the same level and iterate clock. This is a canonical finite Koopman realization. The package does not specify or prove a single family-wide antiunitary that sends every action to its inverse, a nontrivial orbit phase/weight preservation law, or a global self-adjoint Hamiltonian owner. Strict evaluator v0.2 therefore records only **A4_FORMAL_HINT**. The intrinsic number-field tower and Frobenius classes earn the structural-arithmetic verdict at **A0**.

The **A0** audit uses three explicit controls. For neighboring basepoint 3, valuation at primes above 3 and Capelli irreducibility give trivial radical–cyclotomic intersection and the full affine image, so the basepoint-two shared-character cut disappears. The simpler ambient $\text{AGL}_1(\mathbb{Z}/2^n\mathbb{Z})$ control restores exactly 2^{2n-5} elements with four fixed roots. The exact odd-composite ledger below 100 has 25 labels. Its five prime powers 9, 25, 27, 49, 81 are retained as repetition controls attached to the conjugacy classes of Frob_p^r ; they are not new rational-prime Frobenius classes. Only the other 20 mixed composites, each with at least two distinct prime factors, are rejected as having no single-prime Frobenius owner. Empirical density alone receives no **A0** credit.

These **A0** controls do not clear **A1**. The proved affine action and fixed-root law do not include an all-level primitive-cycle/repetition enumeration with completeness control, orbit orientation and phase, multiplicity weights, or monodromy and stability data. Nor do they derive an intrinsic $p \leftrightarrow \gamma_p$, prime-power, or $\log p$ period law. The mandatory shuffled-period, random-weight, random-phase, same-density-length, neighboring-parameter, and simpler-parent tests therefore remain uncompleted at the **A1** orbit layer. Strict evaluator v0.2 records **A1_WEAK**.

There is no cross-prime Fredholm direct sum, no determinant-class regularization, and no target Euler product. We prove no target continuation, functional equation, divisor, counting law, or zero match. Thus both **A2** and **A3** fail. The verdict is **ROUTE_A_EXPLORATORY** under the literal scope **NO_BAD_EULER_OR_ROOT_NUMBER**. The finite permutation unitaries are not a Hilbert–Pólya operator, and Route B is not invoked.

Round 2 advance: inverse-limit, evidence, and Route-A closure. The closed index-two 2-adic image, all-iterate cycle dictionary, dual exact ledgers, the finite Koopman boundary, and explicit target firewall complete the package.

8 Conclusion

The basepoint 2 produces one persistent quadratic entanglement and no further loss: it cuts the affine image by exactly one character at every level. The resulting compatible group law turns a radical-field question into a complete fixed-root and prime-density theorem. The density converges to 7/24, while the character obstruction explains the otherwise surprising absence of exactly four roots. The inverse-limit description is exact, but its arithmetic success does not supply the missing target analytic bridge; the Route-A boundary is therefore part of the theorem package rather than an afterthought.

References

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